

Wireless MIDI Bridge V1
Top-Level System Architecture

Modes

- Wireless BLE MIDI Bridge
- Wired USB passthrough

Power behavior

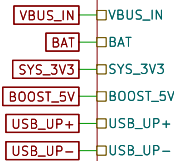
Wireless mode:

- Battery powers system

Passthrough mode:

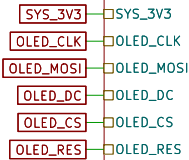
- Upstream USB-C powers system
- Battery does not power system
- Battery charges

Power



Battery/charging/regulators

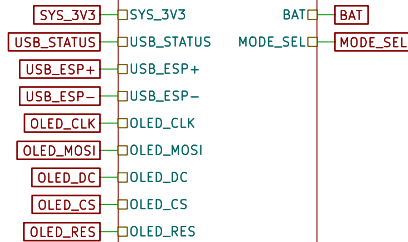
OLED UI



Display + status

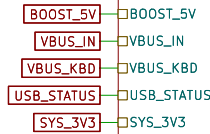
ESP32 inactive in passthrough mode
(no USB host / BLE)

ESP32_S3



BLE + UI + control

VBUS Switching



Keyboard 5V source selection

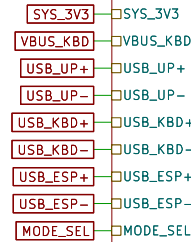
MODE_SEL TRUTH TABLE

USB Data	MODE_SEL = 0	MODE_SEL = 1
	Keyboard <-> Upstream USB-C (direct passthrough)	Keyboard <-> ESP32
BLE	OFF	ON
USB Host	OFF	ON

Mode switching note

Mode switch is authoritative.
Mode switching is not live.
Restart / power cycle required after changing mode.

USB Data



USB mux/port routing

Port definitions

Upstream USB-C:

- Charging input
- Computer connection in passthrough mode

Keyboard USB-C:

- MIDI controller connection
- Routed to ESP32 in wireless mode
- Routed to upstream USB-C in passthrough mode

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Sheet: /
File: midi2BLE.kicad_sch

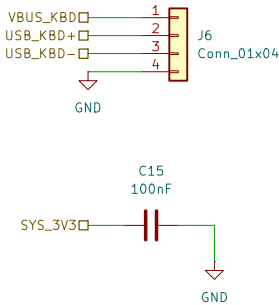
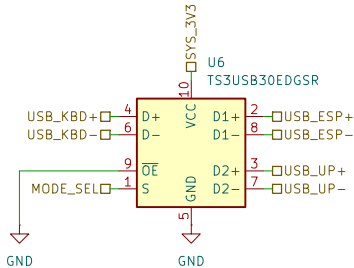
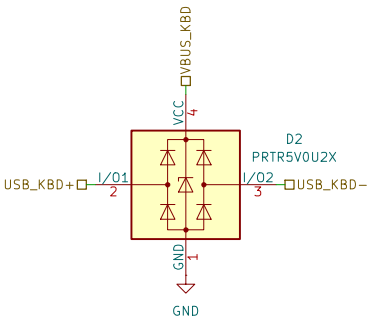
Title: **midi2BLE_Top_Level**

Size: A4 Date: 2026-05-16
KiCad E.D.A. 10.0.0

Rev: **2**
Id: 1/6

Id: 3/6

USB MUX TRUTH TABLE		
	ESP32	USB-C
MIDI_KBD	MODE_SEL OFF	MODE_SEL ON



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Sheet: /USB Data/
File: usb_data.kicad_sch

Title: **midi2BLE_USB_data**

Size: A4 Date: 2026-05-16
KiCad E.D.A. 10.0.0

Rev: **2**
Id: 4/6

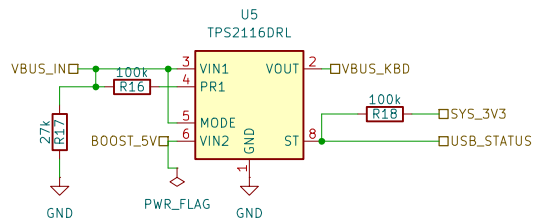
TPS2116 Power Multiplexer
Allows for automatic selection between two 5V sources to power the MIDI keyboard (VBUS_KBD).

VIN1 (VBUS_IN) is given priority. When USB is plugged in, the keyboard is powered directly from the USB 5V rail, preserving battery life and allowing for the battery to charge.

When USB is disconnected, the IC automatically switches to VIN2 (BOOST_5V) which is generated by the MT3608 boost converter from the LiPo battery.

The PR1 resistor divider (R13=100kΩ, R14=27kΩ) sets the switchover threshold at ~4.7V on VIN1. When VBUS_IN drops below this threshold the IC switches to battery power within microseconds with no interruption to the keyboard.

The ST pin outputs the current active source and is routed to ESP32 IO38 so firmware can detect whether the device is USB or battery powered.



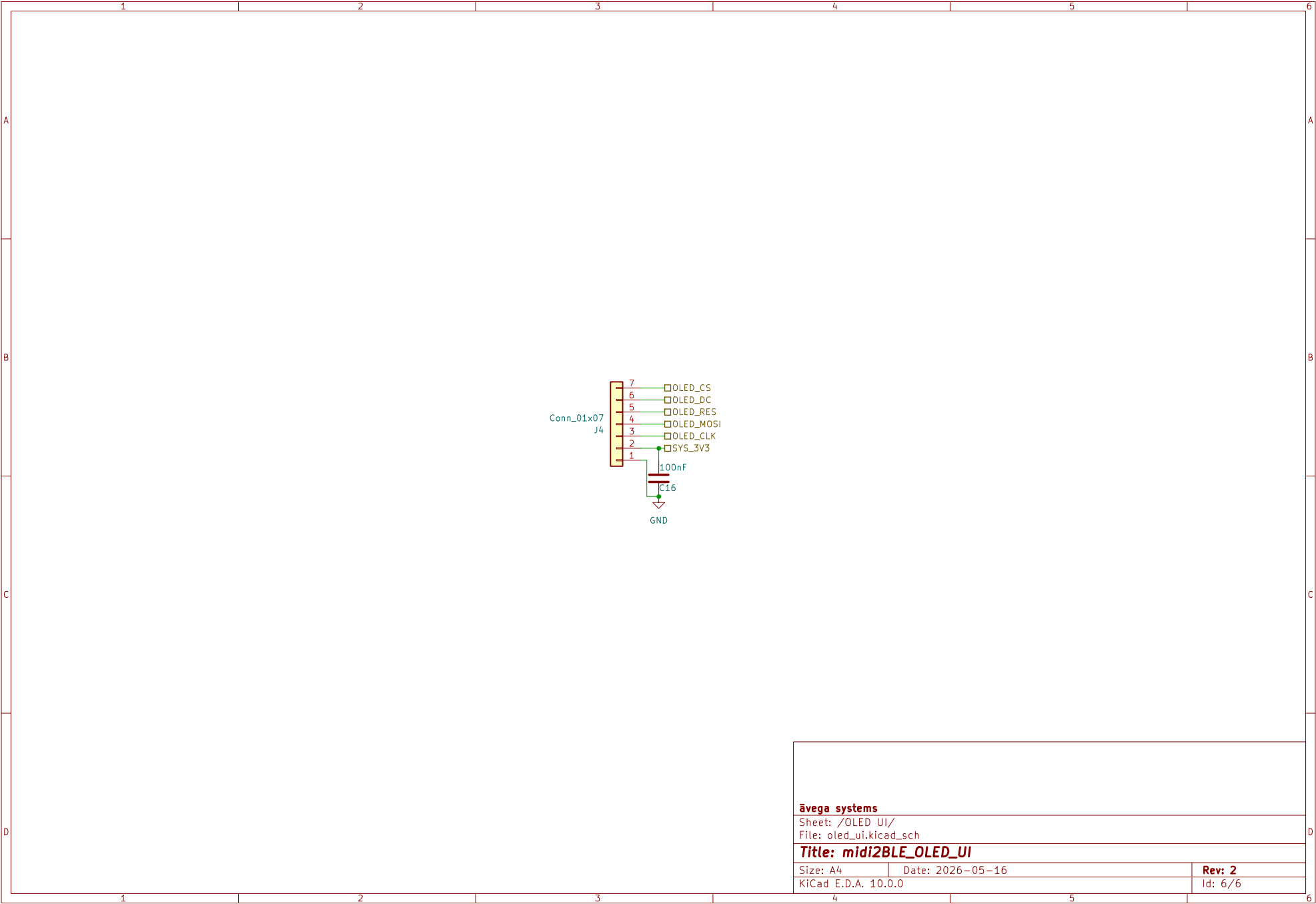
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Sheet: /VBUS Switching/
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Title: midi2BLE_VBUS_Switching

Size: A4 Date: 2026-05-16
KiCad E.D.A. 10.0.0

Rev: 2
Id: 5/6



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Sheet: /OLED UI/
File: oled_ui.kicad_sch

Title: midi2BLE_OLED_UI

Size: A4	Date: 2026-05-16	Rev: 2
KiCad E.D.A. 10.0.0		Id: 6/6